

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

$y < x$

$x < 22$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

Answer

A.

x	y
19	18
20	19
21	20

B.

x	y
19	20
20	21
21	22

C.

x	y
23	22
24	23
25	24

D.

x	y
23	24
24	25
25	26

Correct Answer: A

Rationale

Choice A is correct. The inequality $y < x$ indicates that for any solution to the given system of inequalities, the value of x must be greater than the corresponding value of y . The inequality $x < 22$ indicates that for any solution to the given system of inequalities, the value of x must be less than **22**. Of the given choices, only choice A contains values of x that are each greater than the corresponding value of y and less than **22**. Therefore, for choice A, all the values of x and their corresponding values of y are solutions to the given system of inequalities.

Choice B is incorrect. The values in this table aren't solutions to the inequality $y < x$.

Choice C is incorrect. The values in this table aren't solutions to the inequality $x < 22$.

Choice D is incorrect. The values in this table aren't solutions to the inequality $y < x$ or the inequality $x < 22$.